

Given the following two matrices:

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & i \\ 2 & 0 & 3 \\ 2i & -2i & 2 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 0 & -i \\ 0 & 1 & 0 \\ i & 3 & 2 \end{pmatrix},$$

Using the square matrices as shown above, and the column matrices

$$\mathbf{a} = \begin{pmatrix} i \\ 2i \\ 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ (1-i) \\ 0 \end{pmatrix},$$

find: (a) $\mathbf{Aa}$, (b) $\mathbf{a}^\dagger \mathbf{b}$, (c) $\tilde{\mathbf{a}} \mathbf{Bb}$, (d) $\mathbf{a} \mathbf{b}^\dagger$.

Solution

The square matrices are

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & i \\ 2 & 0 & 3 \\ 2i & -2i & 2 \end{pmatrix} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 2 & 0 & -i \\ 0 & 1 & 0 \\ i & 3 & 2 \end{pmatrix}.$$

Calculate the desired quantities.

$$\mathbf{Aa} = \begin{pmatrix} -1 & 1 & i \\ 2 & 0 & 3 \\ 2i & -2i & 2 \end{pmatrix} \begin{pmatrix} i \\ 2i \\ 2 \end{pmatrix} = \begin{pmatrix} (-1)(i) + (1)(2i) + (i)(2) \\ (2)(i) + (0)(2i) + (3)(2) \\ (2i)(i) + (-2i)(2i) + (2)(2) \end{pmatrix} = \begin{pmatrix} 3i \\ 6 + 2i \\ 6 \end{pmatrix}$$

$$\begin{aligned} \mathbf{a}^\dagger \mathbf{b} &= \begin{pmatrix} i \\ 2i \\ 2 \end{pmatrix}^{*T} \begin{pmatrix} 2 \\ 1-i \\ 0 \end{pmatrix} = \begin{pmatrix} -i \\ -2i \\ 2 \end{pmatrix}^T \begin{pmatrix} 2 \\ 1-i \\ 0 \end{pmatrix} = (-i \quad -2i \quad 2) \begin{pmatrix} 2 \\ 1-i \\ 0 \end{pmatrix} \\ &= ((-i)(2) + (-2i)(1-i) + (2)(0)) \\ &= (-2 - 4i) \end{aligned}$$

$$\begin{aligned} \tilde{\mathbf{a}} \mathbf{Bb} &= \begin{pmatrix} i \\ 2i \\ 2 \end{pmatrix}^T \begin{pmatrix} 2 & 0 & -i \\ 0 & 1 & 0 \\ i & 3 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 1-i \\ 0 \end{pmatrix} = (i \quad 2i \quad 2) \begin{pmatrix} 2 & 0 & -i \\ 0 & 1 & 0 \\ i & 3 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 1-i \\ 0 \end{pmatrix} \\ &= (i \quad 2i \quad 2) \begin{pmatrix} (2)(2) + (0)(1-i) + (-i)(0) \\ (0)(2) + (1)(1-i) + (0)(0) \\ (i)(2) + (3)(1-i) + (2)(0) \end{pmatrix} \\ &= (i \quad 2i \quad 2) \begin{pmatrix} 4 \\ 1-i \\ 3-i \end{pmatrix} \\ &= ((i)(4) + (2i)(1-i) + (2)(3-i)) \\ &= (8 + 4i) \end{aligned}$$

Note that matrix multiplication is associative: $\tilde{\mathbf{a}} \mathbf{Bb} = (\tilde{\mathbf{a}} \mathbf{B}) \mathbf{b} = \tilde{\mathbf{a}} (\mathbf{Bb})$.

Calculate the remaining product.

$$\begin{aligned}\mathbf{a} \mathbf{b}^\dagger &= \begin{pmatrix} i \\ 2i \\ 2 \end{pmatrix} \begin{pmatrix} 2 \\ 1-i \\ 0 \end{pmatrix}^{*T} = \begin{pmatrix} i \\ 2i \\ 2 \end{pmatrix} \begin{pmatrix} 2 \\ 1+i \\ 0 \end{pmatrix}^T = \begin{pmatrix} i \\ 2i \\ 2 \end{pmatrix} \begin{pmatrix} 2 & 1+i & 0 \end{pmatrix} \\ &= \begin{pmatrix} (i)(2) & i(1+i) & (i)(0) \\ (2i)(2) & (2i)(1+i) & (2i)(0) \\ (2)(2) & (2)(1+i) & (2)(0) \end{pmatrix} \\ &= \begin{pmatrix} 2i & -1+i & 0 \\ 4i & -2+2i & 0 \\ 4 & 2+2i & 0 \end{pmatrix}\end{aligned}$$

$\mathbf{a} \mathbf{b}^\dagger$ is the product of a 3×1 matrix and a 1×3 matrix, meaning $\mathbf{a} \mathbf{b}^\dagger$ is a 3×3 matrix.